and capital. Production takes place only when both factors are employed.

- If production has to be equal to one Labour contributes three-fourth of production and capital contributes one-fourth of production.
- The elasticity of substitution between the factors is equal to one.

3.12 Law of Supply

Law of Supply is associated with production analysis. It explains the positive relationship between the price of a commodity and the supply of that commodity. For example, if the price of cloth increases, the supply of cloth will also increase. This is due to the fact that when price rises, it is profitable to increase the production and hence supply increases.

> Law of Supply describes a direct relation between price of a good and the supply of that good.

Definition

The Law of Supply can be stated as:

"*Other things remaining the same, if the price of a commodity increases its quantity supplied increases and if the price of a commodity decreases, quantity supplied also decreases*".

3.12.1 Supply Function

The supply of a commodity depends on the factors such as price of commodity, price of labour, price of capital, the state of technology, number of firms, prices of related goods, and future price expectations and so on. Mathematically the supply function is

$$Q_s = f(P_x, P_r, P_f, T, O, E)$$

Where Q_s = Quantity supplied of x commodity

P_x = Price of x Commodity

P_r = Price of related goods

P_f = Price of factors of production

T = Technology

O = Objective of the producer

E = Expected Price of the commodity.

Assumptions

Law of Supply is based on the following assumptions.

- There is no change in the prices of factors of production
- There is no change in the price of capital goods
- Natural resources and their availability remain the same
- Prices of substitutes are constant
- There is no change in technology
- Climate remains unchanged
- Political situations remain unchanged
- There is no change in tax policy

Explanation

Suppose that the supply function is

$$\mathbf{Q_s = f(P) \text{ or } Q = 20P}$$

P is an independent variable. When its value changes, new values of Q_s can be calculated.

Supply Schedule

A supply schedule shows the different quantities of supply at different prices.

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